8

Recent trends in GNNs-based recommendation

In the previous sections of this monograph, we have primarily focused on GNN-based recommender systems that learn from the user-item interaction graph in its original spatial form, typically without relying on additional sources of information. More recent research has progressively moved beyond this traditional setting by proposing methodologies that either exploit the graph from alternative perspectives or explicitly modify its topology to improve recommendation quality, robustness, or scalability. This section provides a non-exhaustive overview of these emerging directions. We first discuss spectral graph methodologies, highlighting how spectral information can be exploited to partition the interaction graph or filter its structural signals. We then review multimodal and LLM-enhanced recommender systems, showing how external semantic knowledge can be integrated into graph representations and, in the case of LLMs, even used to reshape the topology of the user-item interaction graph itself.

8.1 Spectral graph methodologies

While most GNN-based recommender systems operate in the spatial domain, an alternative line of research adopts a spectral perspective

to analyze and exploit the structural properties of the user-item graph. Rather than focusing solely on message passing over graph neighborhoods, these methods leverage spectral information to improve either the scalability of recommendation algorithms or the quality of the learned representations. In the following, we review two representative research directions, namely, graph partitioning techniques for scalable recommendation and spectral denoising approaches for learning robust user-item representations. Moreover, we present recent studies that empirically investigate how such model families address the topological structure of the user-item graph.

8.1.1 Partition-based recommendation and partitions' topology

The first line of research in spectral-based recommendation addresses the scalability limitations of graph-based recommender systems by explicitly decomposing the recommendation problem into a collection of smaller and more manageable subproblems. Rather than processing the entire item graph as a single structure, these approaches partition it into multiple groups of closely related items, so that similarities can be learned independently within each partition. This divide-and-conquer strategy substantially reduces both computational complexity and memory requirements, making it possible to scale graph-based recommendation to much larger item collections.

In this respect, FPSR (Wei *et al.*, 2023a) follows this principle by first partitioning the item graph into nearly independent subgraphs using graph partitioning techniques. Each partition is processed separately to estimate local item similarities, replacing a single large optimization problem with several smaller ones that can be solved more efficiently and, if desired, in parallel. Although this decomposition greatly improves scalability, partitioning the graph may also remove useful dependencies between items belonging to different subgraphs. To alleviate this issue, FPSR complements the locally learned similarities with a global component derived from the overall structure of the interaction graph, allowing information that spans multiple partitions to be incorporated into the final recommendation model. In this way, the method aims to preserve the benefits of local optimization while retaining a global view

of user-item relationships.

FPSR+ (Wei *et al.*, 2024a) further extends this idea by explicitly addressing situations in which the graph decomposition produces very small or highly unbalanced partitions. In these cases, the limited amount of interaction data available within individual partitions may lead to unreliable similarity estimates. To improve robustness, the method introduces a shared set of representative items that act as bridges across different partitions (defined by the authors as hub items). By allowing these representative items to convey information between otherwise disconnected subgraphs, the model can recover part of the global interaction structure that is lost during partitioning, leading to more stable similarity estimation and improved recommendation.

With reference to these two approaches, the authors in (Gioia *et al.*, 2026a) have recently investigated the sensitivity of the FPSR family to the graph partitioning strategy by analyzing the impact of the partition's topology (in terms of its granularity) on recommendation performance. Particular attention is devoted to the partition-size parameter, which determines how granular the item graph is decomposed. Since the effectiveness of graph partitioning depends on balancing computational efficiency with the preservation of meaningful structural information, the authors examine how different partition sizes affect the behavior of both the original FPSR and other variants of the FPSR+ version, namely, FPSR+$_D$ and FPSR+$_F$.

The results reported in Table 8.1 on the Douban dataset (Shi *et al.*, 2015) show that the partition size (indicated as τ) has a substantial influence on the performance of the original FPSR model. When the graph is divided into excessively small partitions, recommendation accuracy deteriorates noticeably because each partition contains only limited interaction information and captures fewer relationships with items belonging to other partitions. As the partition size increases reaching the best found partition size (τ_{best}), this effect is progressively mitigated, leading to significantly more accurate recommendations.

FPSR+$_D$ and FPSR+$_F$ exhibit a considerably more stable behavior across different partitioning settings. By introducing representative items that are shared across partitions, these models preserve part of the global interaction structure that would otherwise be lost during

Table 8.1: Recommendation results of FPSR, FPSR+$_D$, and FPSR+$_F$ with varying values of τ (i.e., partition size) on Douban, as presented in the work by Gioia *et al.* (2026a). The value τ_{best} stands for the best configuration found in the authors' analysis.

	FPSR ($\tau_{\text{best}} = 0.5$)		FPSR+$_D$ ($\tau_{\text{best}} = 0.4$)		FPSR+$_F$ ($\tau_{\text{best}} = 0.4$)	
	R@20	N@20	R@20	N@20	R@20	N@20
$\tau = 0.05$	0.1631	0.1561	0.2068	0.1882	0.1731	0.1602
$\tau = 0.15$	0.1918	0.1767	0.2097	0.1908	0.1953	0.1771
$\tau = 0.25$	0.1930	0.1780	0.2109	0.1912	0.1957	0.1780
τ_{best}	0.2034	0.1855	0.2132	0.1928	0.2041	0.1841

graph decomposition. This mechanism reduces the sensitivity of local similarity estimation to small or unbalanced partitions, resulting in more consistent recommendation performance even under aggressive partitioning strategies. Among the two variants, FPSR+$_D$ demonstrates the highest degree of robustness, whereas FPSR+$_F$ provides a more moderate but still noticeable improvement over the original FPSR.

Overall, the analysis highlights that the effectiveness of partition-based recommendation depends not only on the graph decomposition itself but also on how information is propagated across partition boundaries. Appropriate tuning of the partition granularity can substantially improve the performance of the original FPSR model, while the introduction of hub items in FPSR+ offers an additional level of robustness by preserving global structural information when the graph is partitioned into many small subgraphs.

8.1.2 Robustness of spectral denoising methods to data sparsity

Rather than processing the interaction graph exclusively in the spatial domain, spectral denoising methods analyze its spectral structure to identify the most informative patterns while attenuating noisy or redundant signals. The underlying intuition is that user preferences often exhibit an inherent low-dimensional organization, meaning that much of the relevant information can be captured by a limited number of dominant latent components.

Building on this intuition, these methods first derive a compact

spectral representation of the interaction graph and then exploit it in different ways. For instance, SVD-GCN (Peng *et al.*, 2022b), already discussed in this monograph, constructs an initial low-rank representation of the user-item graph and subsequently refines it through graph convolutional layers. Other models further extend this idea by explicitly manipulating the spectral representation. SGDE and RSGDE (Peng *et al.*, 2024b) selectively suppress or reweight less informative spectral components to reduce the influence of noise and alleviate issues such as over-smoothing, while CSGDE (Peng *et al.*, 2024b) complements this strategy with a contrastive learning objective aimed at preserving discriminative information in the learned embeddings. A different direction is followed by SGFCF (Peng *et al.*, 2024a), which applies a predefined spectral filter to the interaction graph, producing recommendation scores through a closed-form solution without requiring iterative optimization.

Although these models differ in their architectural choices and training strategies, they all share a common view of recommendation as a spectral filtering problem. Their main objective is to preserve the dominant structural patterns of the user-item interaction graph while suppressing components that are more likely to reflect noise or spurious correlations. Consequently, the recommendation process can be interpreted as enhancing the most informative spectral signals rather than treating all graph components as equally relevant.

The effectiveness of this family of methods relies on the assumption that the dominant spectral components of the interaction graph encode the essential preference patterns shared across users and items, whereas weaker components predominantly capture noise or less informative variations. Under this perspective, emphasizing the most significant spectral structures leads to cleaner representations and, ultimately, to more reliable recommendations even in data sparse scenarios.

To investigate whether spectral denoising methods are intrinsically more robust to sparse interaction data, the authors in (Gioia *et al.*, 2026b) have recently conducted a controlled study on the Douban dataset (Shi *et al.*, 2015), where the amount of training information is progressively reduced while keeping the evaluation protocol unchanged. This setting provides a convenient testbed for analyzing the behavior of

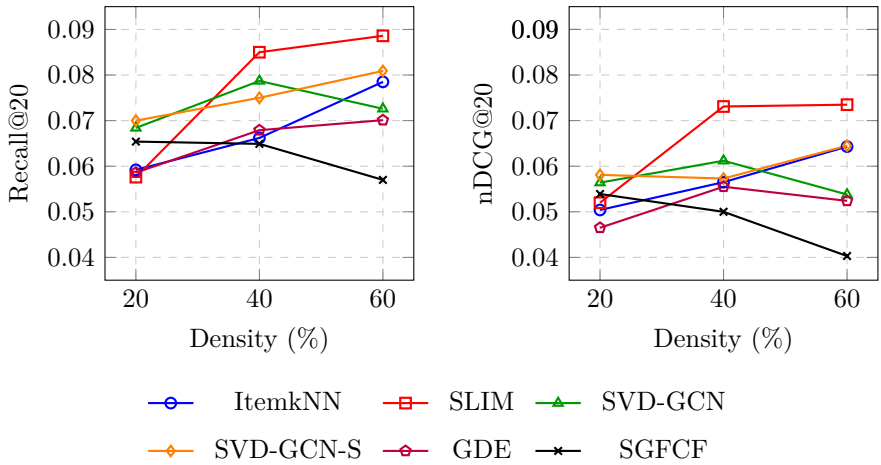


Figure 8.1: Recommendation performance on the Douban dataset when considering different density levels (20%, 40% and 60%) on the training set. Results are taken from the work by Gioia *et al.* (2026b).

spectral denoising models under increasing levels of sparsity without introducing additional sources of variability.

As illustrated in Figure 8.1, the analysis focuses on representative methods from the spectral denoising family, namely SVD-GCN, SVD-GCN-S (a variant of SVD-GCN), GDE (Peng *et al.*, 2022a), and SGFCF. Their performance is compared against two well-established neighborhood-based recommenders, SLIM (Ning and Karypis, 2011) and ItemkNN (Deshpande and Karypis, 2004), which serve as strong non-spectral baselines.

To simulate increasingly sparse scenarios, the test set is kept fixed while only the amount of training data is reduced. After partitioning the dataset into training, validation, and test sets, progressively smaller training sets are generated by retaining only 60%, 40%, and 20% of the original training interactions. This protocol isolates the effect of diminishing training signal, ensuring that any variation in performance can be attributed solely to the increasing sparsity of the available interaction data.

The results reported in Figure 8.1 indicate that the performance of nearly all methods deteriorates as fewer training interactions become

available. Among the spectral approaches, SGFCF exhibits a distinctive behavior, maintaining stable performance and even showing a slight improvement under the sparsest setting, suggesting that its fixed spectral filtering strategy is less sensitive to the reduction of training data than the mechanisms adopted by the other models. In contrast, SVD-GCN, SVD-GCN-S, and GDE follow a degradation trend comparable to that observed for the non-spectral baselines. Although SVD-GCN and SVD-GCN-S remain competitive in the most challenging setting, they do not consistently outperform SLIM or ItemkNN, while GDE does not display any evident robustness advantage.

Overall, these findings suggest that emphasizing the dominant spectral components of the interaction graph does not automatically translate into greater robustness to data sparsity. Instead, the benefits appear to depend on the specific filtering strategy adopted by each model, with SGFCF representing a notable exception. More generally, the experimental evidence indicates that robustness to sparse interaction data should be regarded as an empirical property of individual model-dataset combinations rather than as an inherent consequence of spectral denoising itself.

8.2 Multimodality and large language models

Recent advances in graph-based recommender systems have progressively moved beyond the sole exploitation of user-item interactions by incorporating complementary sources of information. In particular, multimodal data and large language models (LLMs) have emerged as two major research directions for enriching graph representations with semantic knowledge that is not directly observable from interaction data alone. In the following, we first provide an overview of multimodal GNN-based recommendation and then focus on LLM-enhanced recommender systems, highlighting how LLMs can explicitly reshape the topology of the user-item interaction graph. We deepen into this aspect through a recent empirical analysis that investigates how LLMRec (Wei *et al.*, 2024b), a notable example in the literature of multimodal LLM-based recommendation, modifies the topology of the user-item graph.

8.2.1 Multimodal GNNs-based recommendation: an overview

The integration of multimodal information into recommender systems (Liu *et al.*, 2025; Liu *et al.*, 2024; Malitesta *et al.*, 2025; Pan *et al.*, 2026; Xu *et al.*, 2025b; Zhou *et al.*, 2023a) has demonstrated substantial benefits across a wide range of application domains, including e-commerce (Xv *et al.*, 2023), fashion (Chen *et al.*, 2019), music (Oramas *et al.*, 2017), food (Min *et al.*, 2020), and micro-video recommendation (Cai *et al.*, 2022). By leveraging multiple data modalities associated with items, these systems are able to learn richer representations that improve recommendation quality. Recent advances in multimodal learning (Baltrusaitis *et al.*, 2018) have further accelerated the development of multimodal recommender systems, enabling them to tackle long-standing issues such as data sparsity and the cold-start problem (He and McAuley, 2016b; Oramas *et al.*, 2017). Moreover, multimodal information facilitates a deeper understanding of user-item interactions by exploiting attention mechanisms, which can reveal users' preferences and intentions while also improving the interpretability of recommendation models (Chen *et al.*, 2019).

The rapid adoption of graph neural networks in recommender systems (He *et al.*, 2020) has encouraged the development of methods that incorporate multimodal information into both user-item bipartite graphs and knowledge graphs (Wang *et al.*, 2023). These approaches typically refine user and item representations through message-passing mechanisms that effectively propagate multimodal information across the graph structure. Early graph-based methods primarily relied on injecting multimodal item features into the recommendation pipeline (Wei *et al.*, 2019). More recent solutions, however, construct dedicated graph representations for each modality and explicitly disentangle user preferences across different modalities (Kim *et al.*, 2022), while simultaneously modeling both global and local user interests (Guo *et al.*, 2024). Other studies focus on capturing the structural relationships among multimodal item representations within the catalog (Liu *et al.*, 2021; Zhang *et al.*, 2021; Zhou *et al.*, 2023b; Zhou and Shen, 2023), often by incorporating self-supervised (Tao *et al.*, 2023; Wei *et al.*, 2023b; Xu *et al.*, 2025a; Yu *et al.*, 2023; Zhou *et al.*, 2023c) and contrastive (Su *et al.*,

2024; Yi *et al.*, 2022) learning paradigms. More recently, researchers have explored the integration of advanced generative models, including variational autoencoders (Yi and Chen, 2024; Zhou and Miao, 2024) and diffusion models (Jiang *et al.*, 2024a). Additional research directions include transforming multimodal representations into the frequency domain (Ong and Khong, 2025; Yu *et al.*, 2025), transferring multimodal knowledge from large-scale architectures to more compact models (Wang *et al.*, 2024a; Wei *et al.*, 2024c), and improving recommendations for long-tail items in the catalog (Lin *et al.*, 2024).

8.2.2 Multimodal LLMs & GNNs: the notable example of LLMRec

A representative example of the integration of large language models (LLMs) and multimodal information into graph-based recommender systems is LLMRec (Wei *et al.*, 2024b), one of the earliest frameworks to combine these paradigms within a unified recommendation architecture. LLMRec enhances graph-based recommendation through LLM-driven graph augmentation and dedicated mechanisms for denoising and feature refinement. By exploiting the semantic reasoning capabilities and extensive knowledge encoded in LLMs, LLMRec alleviates the limitations imposed by sparse implicit feedback, ultimately improving the quality of user-item representations. More specifically, the interaction graph is enriched by augmenting three complementary components: (i) user-item interaction edges; (ii) item node attributes; and (iii) user node profiles. For the sake of this monograph, we only deepen into the description of the graph and node augmentation procedures, leaving aside the presentation of the denoising mechanisms; the interested reader may refer to the original work for a complete presentation of the approach.

To augment the user-item interaction graph, LLMRec employs the LLM as a knowledge-aware sampler that generates informative pairwise user-item training instances. This strategy aims to alleviate the effects of noisy interactions, reducing both false-positive user-item relationships and false-negative non-interactions. For each user, the LLM is prompted with the historical interaction sequence, the corresponding side information (e.g., year and genre), and a candidate item set produced by a backbone recommender such as LATTICE (Zhang *et al.*,

2021) or MMSSL (Wei *et al.*, 2023b). Based on this information, the LLM identifies the items that the user is most likely and least likely to interact with. These predictions are then adopted as positive and negative samples during the BPR optimization (Rendle *et al.*, 2009).

The framework also exploits the LLM to augment node attributes. Specifically, user profiles are generated by summarizing each user's historical interactions, whereas enriched item descriptions are produced from prompts containing the available item metadata. The resulting textual information is subsequently encoded into embedding representations and incorporated into the recommendation model.

Once the augmented user and item information has been generated, LLMRec incorporates it into the original graph through a three-stage pipeline. First, an augmented semantic projection module reduces the dimensionality of the LLM-enhanced feature vectors. Next, a collaborative context injection stage exploits a lightweight Graph Neural Network (GNN) encoder to infuse higher-order collaborative signals into the projected user and item features. Finally, a semantic feature fusion module combines the enhanced semantic representations with the corresponding collaborative embeddings using a weighting coefficient and L_2 normalization, yielding the final user and item representations. Notably, the model is optimized using triplets sampled from the union of the original user-item graph set and a subset of the LLM-augmented interactions.

8.2.3 A topological analysis of LLMRec's augmentation strategy

One of the main strategies representing the key novelty of LLMRec is the augmentation of the historical user-item interaction graph through LLMs to refine noisy recorded edges among users and items. In this respect, it would be interesting to understand how the same graph changes after the LLM intervention from a *topological* viewpoint. This is, in fact, an open research question, not necessarily referred to the graph-based recommendation domain. Other works, where the most prominent is the one by Fatemi *et al.* (2024), have already questioned how and to what extent LLMs are able to understand and act over graph data. In this sense, the recent work by Fioretti *et al.* (2025)

analyzes this research dimension for the prominent example of LLMRec, studying the topological impact of the LLM's augmentation on the user-item interaction graph.

Table 8.2 summarizes the topological analysis conducted by Fioretti *et al.* (2025) on the Amazon-Music (He and McAuley, 2016a; McAuley *et al.*, 2015) and the Netflix dataset[1]; as for the Netflix dataset, the authors consider two versions (indicated in the table with Netflix$^{(1)}$ and Netflix$^{(2)}$), namely, with standard and a custom training splitting. The authors' rationale is to compare the original interaction graphs with those obtained through LLM-based augmentation. The experiments consider two graph-based recommenders, LATTICE and MMSSL, as candidate generators, and three LLMs (*gpt-3.5-turbo-16k*[2], *gpt-4-turbo*[3], and *Llama-3-405B-Instruct*[4]) to refine the candidate items.

Overall, the reported results consistently indicate that LLM-based augmentation substantially reshapes the topology of the user-item graph. The most evident effect is the increase in both Density and Average Degree across all datasets and experimental configurations, suggesting that the augmented graphs become more connected and contain a richer set of interactions. Such an increase is observed for both user and item partitions, indicating that users are connected to a larger number of items while items receive interactions from a broader set of users. According to the authors, this denser connectivity may facilitate information propagation through the graph and ultimately provide a richer space for recommendation.

The degree distributions also exhibit noticeable changes after augmentation. In particular, the Gini coefficient computed on the user partition consistently decreases, indicating that user activity becomes more evenly distributed rather than being concentrated on a limited number of highly active users. Conversely, the Gini coefficient on the item partition generally shows a slight increase, especially when MMSSL is adopted as the backbone recommender, suggesting that item popularity remains relatively concentrated and may even become slightly more

[1] Shortened link to the dataset: https://tinyurl.com/3wk3xchb.
[2] https://developers.openai.com/api/docs/models/gpt-3.5-turbo.
[3] https://developers.openai.com/api/docs/models/gpt-4-turbo.
[4] https://huggingface.co/meta-llama/Llama-3.1-405B-Instruct.

Table 8.2: Topological analysis of LLMRec's augmentation strategy on the user-item graph, considering different LLMs and recommendation backbones. Results are taken from the study by (Fioretti *et al.*, 2025).

LLM	Baseline	Density	Gini-U	Gini-I	AvgDeg-U	AvgDeg-I	AvgClustC-U	AvgClustC-I	Assort-U	Assort-I
					Amazon-Music					
Original	–	0.0028	0.4929	0.5012	5.6015	14.4962	0.1184	0.0496	-0.0884	0.0533
gpt-3.5-turbo-16k	LATTICE	0.0038	0.3632	0.5497	6.3730	16.4927	0.1117	0.0470	-0.0371	0.0322
	MMSSL	0.0038	0.3632	0.5780	6.6823	17.2932	0.1114	0.0457	-0.0164	0.0333
					Netflix[(1)]					
Original	–	0.0006	0.1187	0.7493	5.2280	8.3996	0.1102	0.1474	0.2250	-0.0357
gpt-3.5-turbo-16k	LATTICE	0.0009	0.0859	0.7839	6.3641	10.2006	0.0963	0.1387	0.1671†	-0.1542
	MMSSL	0.0009	0.0859	0.8076	6.3907	10.2554	0.0956	0.1285	0.2529†	-0.1857
					Netflix[(2)]					
Original	–	0.0006	0.1467	0.7341	4.2291	7.4407	0.1380	0.1605	0.2502	-0.0054
Llama-3-405B-Instruct	LATTICE	0.0008	0.0997	0.7994	6.2271	10.9053	0.1019	0.1261	0.2001†	-0.1710
	MMSSL	0.0008	0.0997	0.8147	6.2265	10.9042	0.1026	0.1101	0.2738†	-0.2408
gpt-4-turbo	LATTICE	0.0008	0.1037	0.7977	6.1347	10.7434	0.1037	0.1276	0.1506†	-0.1643
	MMSSL	0.0008	0.1033	0.8120	6.0657	10.6227	0.1044	0.1143	0.2048†	-0.2425

[(1)] With the original training set by the authors.
[(2)] With a 90-10 random split of the original training set.
† These values represent a trend divergence with respect to the original dataset version.

skewed despite the richer interaction graph.

The structural organization of the graph is further characterized by a moderate reduction in the Average Clustering Coefficient for both user and item partitions, with the decrease being more pronounced on the item side. Since this metric captures the tendency of nodes to share common neighborhoods, the observed reduction suggests that the newly introduced interactions are distributed across a broader portion of the graph instead of reinforcing existing local communities. The authors interpret this as an indication of increased interaction diversity, which could positively affect the diversity of the generated recommendations.

Finally, the analysis of Degree Assortativity reveals more dataset-dependent trends. On the Amazon-Music dataset, User Assortativity increases while remaining negative, indicating a partial reduction of the graph's disassortative structure, whereas Item Assortativity decreases while remaining positive, implying that highly popular items become less likely to be associated with other popular items. On both Netflix dataset versions, User Assortativity remains positive, although the direction and magnitude of the variation depend on the specific combination of backbone recommender and LLM. This is the only topological property for which the authors observe inconsistent trends across all experimental settings, a behavior highlighted in Table 8.2 through the dagger (†)

notation. In contrast, Item Assortativity consistently becomes more negative, suggesting stronger interactions between highly popular and less popular items. Moreover, employing MMSSL as the generator generally produces larger absolute assortativity values than LATTICE.